PAPER_475 — UQFF Sub-Term Physics Modules Catalogue (44 Standalone Calculators)

Author: Daniel T. Murphy

Abstract

This whitepaper catalogues 44 standalone sub-term physics calculator modules that constitute the atomic building blocks of the UQFF and MUGE frameworks. Each module encapsulates a single physical quantity — from aether vacuum densities and buoyancy coupling constants to solar cycle frequencies and stellar surface temperatures — as a self-contained C++ class. This catalogue provides a reference index for module equations, key parameter values, output ranges, and integration targets within MAIN_1_CoAnQi.cpp and CondensedPhysics.py.

---

1. Introduction

The 44 sub-term modules were extracted from `grok_share_b0a3dc1d.txt` (10,420 lines) and represent the granular physics underpinning the UQFF unified field equation:

$$F_{U,Bi,i} = \int \left[\text{Ug coupling} \cdot \text{vacuum terms} \cdot \text{buoyancy} \cdot \text{aether} \cdot \text{stellar params}\right] dx$$

Each module is implemented as a C++ class in `modules/subterms/` with standard interface: `computeX()`, `updateVariable()`, `getEquationText()`, `printVariables()`.

---

2. Module Catalogue

Category A: Vacuum Energy Modules

#	Module	Key Equation	Key Value
1	`AetherVacuumDensityModule`	ρ_vac_A = E_A/c^2	7.09×10^{-36} J/m^3
2	`UniversalInertiaVacuumModule`	ρ_vac_UA [energy]	7.09×10^{-36} J/m^3
3	`ScmVacuumDensityModule`	ρ_vac_SCm = ρ_UA/10	7.09×10^{-37} J/m^3
4	`UaVacuumDensityModule`	ρ_UA = E_UA/c^2 [mass]	7.88×10^{-53} kg/m^3
5	`BackgroundAetherModule`	A_μ = (ρ_A/c^2)∂_$\mu\varphi$	ρ_A = 7.09×10^{-36} J/m^3

Category B: Coupling Constants

#	Module	Key Equation	Key Value
6	`AetherCouplingModule`	A_$\mu\nu$ = g_$\mu\nu$ + η T_s^$\mu\nu$	$\eta \approx$ 1/E_s_total $\approx$ 1e-15

#	Module	Key Equation	Key Value
7	UgCouplingModule	k_i weights Ug_i	k_1=1.5, k_2=1.2, k_3=1.8, k_4=1.0
8	BuoyancyCouplingModule	U_bi = -β_i U_gi Ω_g...	β_i = 0.6 uniform
9	InertiaCouplingModule	I_e = M r^2 Ω^2/c^2	Sun: $\approx 7.8\times10^{-10}$
10	UgIndexModule	Ug_{i,n} = G M/r^2 Q_i...	4×26 array

Category C: Solar/Stellar Parameters

#	Module	Key Equation	Key Value
11	SolarWindBuoyancyModule	ε_sw = 0.001 × (1 + A sin(ωt))	ε_sw = 0.001 baseline
12	SolarWindModulationModule	v_sw(t) = v_0(1+A sin(ωt))	v_0=400 km/s, A=0.2
13	SolarWindVelocityModule	v_sw $\in$ [400,800] km/s	v_fast = 750 km/s
14	SolarCycleFrequencyModule	f_sc = 1/(11 yr)	2.88×10^{-9} Hz
15	HeliosphereThicknessModule	L = r_HP - r_TS	~30 AU (~4.5×10^{12} m)
16	StellarMassModule	M_s(t) = M_0(1-γ_ML t)	γ_ML $\approx 1.4\times10^{-14}$ s^{-1}
17	StellarRotationModule	ω_s = 2π/P_rot	Sun: 2.87×10^{-6} rad/s
18	SurfaceMagneticFieldModule	B_s = μ_0 M_mag/4π r^3	Magnetar: 4.4×10^{13} T
19	SurfaceTemperatureModule	T_s = (L/4$\pi\sigma r^2$)^0.25	Sun: 5778 K
20	MagneticMomentModule	μ = I A_vort	Solar: ~1×10^{21} A·m^2

Category D: Galactic/Astrophysical Parameters

#	Module	Key Equation	Key Value
21	GalacticDistanceModule	d_g = virial radius	MW: 2.55×10^{20} m
22	GalacticSpinModule	Ω_g = 2π/T_gal	MW: 7.3×10^{-16} rad/s
23	GalacticBlackHoleModule	M_BH $\propto \sigma^4$ (M-σ)	Sag A*: 8×10^{36} kg
24	FeedbackFactorModule	F_env = f_AGN+f_SN+f_SF	f_AGN=0.1, f_SN=0.05
25	MagneticStringModule	T_s = (μ_0 I^2/4π) ln(L/a)	~10^{28} N (cosmic string)
26	Ug3DiskVectorModule	Ug3_disk = G M_disk/r^2(h/r)	MW: h/r $\approx$ 0.07
27	Ug1DefectModule	Ug1_corr = Ug1(1-δ_def)	δ_def $\approx$ 0.05-0.15

Category E: Quantum & Field Calculators

#	Module	Key Equation	Key Value
28	DPMModule	26-sphere: $(x-h)^2+...=r^2$	SCm E = 10^{42} J
29	UnifiedFieldModule	F_U = Σ k_i Ug_i + F_bi...	Orchestrator
30	StressEnergyTensorModule	T_$\mu\nu$ = (ρ+p)u_μu_ν+pg_$\mu\nu$	Trace = -ρ+3p
31	QuasiLongitudinalModule	E_QL = ε_0 E^2/2	~1e-12 J/m^3
32	OuterFieldBubbleModule	r = r_0 exp(H t)	At 10 Gyr: r = 1.28 r_0
33	ReciprocationDecayModule	γ = γ_0 exp(-t/τ_rec)	τ_rec ~ 1 Gyr
34	ScmPenetrationModule	δ_SCm = λ(ρ/ρ_SCm)^0.5	London-analog depth
35	ScmReactivityDecayModule	d[SCm]/dt = -k_r [SCm]	[SSq]=0.57, k_r~1e-18 s^{-1}
36	ScmVelocityModule	v_SCm = c/n_SCm	n_SCm $\approx$ 1.0000001
37	PiConstantModule	π = 4Σ(-1)^k/(2k+1)	Leibniz series
38	CorePenetrationModule	δ = (ρ_core/ρ_avg)^n r_core	NS core: $\delta \to 0$
39	NegativeTimeModule	g(t<0) = g_0 cos(ωt)	f_TRZ = 0.1

#	Module	Key Equation	Key Value
40	`TimeReversalZoneModule`	f_TRZ = r_TRZ/r	Canonical: 0.1
41	`HeavisideFractionModule`	H(f) = 0 (f<0), 1 (f≥0)	Threshold: 0
42	`StepFunctionModule`	θ(x): boxcar windows	Regime switching

Category F: Index / Utility Modules

#	Module	Key Equation	Key Value
43	`GalacticBlackHoleModule`	g_BH = G M_BH/r^2	At r=5.5×10^{10} m (Sag A*)
44	`InertiaCouplingModule`	γ_I = 1 + I_c	Ug3 relativistic boost

---

3. Integration Targets

3.1 MAIN_1_CoAnQi.cpp

Sub-term modules are candidates for batch extraction as `PhysicsTerm` subclasses:

```
// Example: ScmVacuumDensityModule → SCmVacuumDensityTerm
class SCmVacuumDensityTerm : public PhysicsTerm {
public:
    double compute(const SystemParams& params) override {
        return 7.09e-37; // ρ_vac_SCm [J/m³]
    }
    std::string getName() const override { return "SCm Vacuum Density"; }
};
```

Target integration point: **Batch 24** in SOURCE1-116 registry near line 6688+.

3.2 CondensedPhysics.py

Each sub-term module maps to a `Calculator` class method:

```
class SubTermCalculator:
    def compute_aether_vacuum_density(self) -> dict:
        # Returns ρ_vac_A = 7.09e-36 J/m³ with equation text
```

Target: new `UQFFSubTermCalculator` in CondensedPhysics.py Part 5 (after line 60000).

3.3 index.js

Module constants exportable as:

```
const SUB_TERMS = {
  rho_vac_UA: 7.09e-36,     // J/m³
  rho_vac_SCm: 7.09e-37,    // J/m³
  beta_i: 0.6,               // buoyancy coupling
  kappa: 0.0005/86400,       // per-second DPM rate
  SSq: 0.57,                 // SCm calibration
};
```

---

4. Calibration Constants Summary

Constant	Value	Source Module
κ	0.0005/day	DPMModule
[SSq]	0.57	ScmReactivityDecayModule
β_i	0.6 (all i)	BuoyancyCouplingModule
ε_sw	0.001	SolarWindBuoyancyModule
η	~10^{-15}	AetherCouplingModule
H_SCm	0.99	ScmVacuumDensityModule
k_1,k_2,k_3,k_4	1.5/1.2/1.8/1.0	UgCouplingModule
f_TRZ	0.1	TimeReversalZoneModule

---

5. Scientific Context

The 44 sub-term modules collectively represent a field-theoretic decomposition of gravity that accounts for:

- **Vacuum energy hierarchy**: ρ_UA : ρ_SCm : ρ_cosm = 10 : 1 : 0.001
- **Coupling hierarchy**: k_3 (string/rotation) > k_1 (dipole) > k_2 (charge) > k_4 (vacuum)
- **Time evolution**: DPM birth model $\rightarrow$ inflation $\rightarrow$ SCm decay $\rightarrow$ solar wind modulation $\rightarrow$ present
- **Multi-scale validity**: Sub-parsec (heliosphere) to Gpc (galaxy clusters)

---

6. Conclusion

This catalogue documents 44 atomic physics calculator modules that implement the sub-term physics of the UQFF/MUGE framework. The modules span vacuum energy, coupling constants, stellar parameters, galactic scales, and quantum field calculators. They constitute the `modules/subterms/` library and are ready for integration as `PhysicsTerm` subclasses in MAIN_1_CoAnQi.cpp or as Python calculator methods in CondensedPhysics.py.

---

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_\mu \phi_{\text{NS}})(\partial^\mu \phi_{\text{NS}}) - V(\phi_{\text{NS}}) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac,[SCm]}} \cdot f_{\text{SCm}} \cdot (1-e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\text{NS}}) = \frac{1}{2}m^2\phi_{\text{NS}}^2 + \frac{\lambda}{4!}\phi_{\text{NS}}^4 + \kappa \cdot \rho_{\text{vac,[SCm]}} \cdot \phi_{\text{NS}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\text{NS}}} = \nabla^2\phi_{\text{NS}} - (4\pi G\rho_{\text{NS}}/c^2)\phi_{\text{NS}} + \Omega_{\text{spin}}\partial_t\phi_{\text{NS}} = 0}$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \xrightarrow{\text{DPM + ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{\text{4 forces}} F_{U\_Bi\_i} \xrightarrow{\text{sector E-L}} \delta S/\delta\phi_{\text{NS}} =$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

---

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac,[SCm]}}/\rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac,[SCm]}} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\text{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.106 (near-threshold regime), placing it in the $t \to \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36}$ kg/m^3.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 101, \quad n_{\text{channel}} = 8/26$$

Since $p_{\text{DVP}} = 101$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair $(f'_{\text{UA}} + f_{\text{SCm}} = 1)$ constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is $\mathbf{10^4}$ **yr** (spin-down equilibrium):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq]\cdot m/M_\odot}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.106	✓ Threshold-consistent
DVP prime	$p_k \in$ {2,3,...,113}	$p_{\text{DVP}} = 101$	✓ Resonant
BSH layers	26 harmonic terms	j = 1...26, $\cos(2\pi j/26)$	✓ Full 26D projection
κ decay	5.0×10^{-4} day^{-1}	Applied in VDS exponential	✓ Canonical
[SSq]	0.57	Applied in BSH saturation	✓ Canonical

---

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Thomson σ_T (QED synchrotron)	UQFF U_m scattering kernel: σ_T = 6.6524×10^{-29} m^2	σ_T = 6.6524×10^{-29} m^2 (PDG QED exact)	PDG 2024	100% (exact QED input)
Astrophysical system luminosity X-ray / Radio	UQFF MUGE g_total $\rightarrow$ L_X via Stefan-Boltzmann + buoyancy flux: L_X $\approx$ g_total × M_env	L_X L $\geq 10^{37}$ erg/s	Chandra CXC	✓ Consistent order of magnitude
GR Schwarzschild limit	UQFF g_total must satisfy g $\leq$ c^2/(2r_s) at event horizon	r_s = 2GM/c^2 (GR exact)	PDG 2024 / GR	✓ UQFF respects GR horizon
κ vacuum rate vs X-ray variability	UQFF κ = 0.0005/day $\rightarrow$ timescale τ_UQFF = 2000 days	Observed X-ray variability τ_obs (instrument monitoring)	Chandra CXC	Testable UQFF variability timescale

New physics claim: UQFF MUGE generates gravity enhancement factors (g_total/g_Newt > 1) for Astrophysical system through vacuum buoyancy coupling — a mechanism absent from GR+SM. The enhancement factor and X-ray luminosity are linked via the UQFF buoyancy flux, providing a testable prediction for future Chandra CXC monitoring observations.

Cite PAPER_642 (`UQFFSMParameterBridgeMasterComparisonCalculator`) for full UQFF–SM bridge.

Source: `grok_share_b0a3dc1d.txt` L1502–9356 (44 class definitions)
Header index: `modules/subterms/sub_terms_index.h`
Tags: sub-terms, catalogue, vacuum-density, coupling-constants, solar-parameters, UQFF, MUGE, integration

---

Appendix: Session 204 Codebase Upgrade Reference

> *Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by `upgrade_kozima_ramanujan_appendices.py`. For detailed derivations, see PAPER_840/851/852/855.*

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
`fneutron_s26_coupling.py`	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
`kozima_scm_cross_section.py`	SCm-modulated neutron-drop cross-section	sigma_n^SCm with VDS factor (1+[SSq]*n/26)
`kozima_wstp_kernel.py`	11-symbol Wolfram export (`UQFFKozima`)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: F_neutron^SCm = N_n * sigma_n^SCm(omega) * Phi_phonon * (F_{U,Bi}/F_U - 1) where sigma_n^SCm(omega,n) = sigma_0 * exp[-(omega-omega_SCm)$^{2/(2*\text{Gamma}}$2)] * (1 + [SSq]*n/26)

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
`ramanujan_polylog_s26.py`	Li_26([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
`s26_wstp_kernel.py`	8-symbol Wolfram export (`UQFFS26`)	S26, R26, NaiveLi, S26VDS

Core equation: S_26(z) = Li_26(z) = eta_26(z)/(1-2^{1-26}) + 2$^{\{1\text{-}26\}/(1\text{-}2}${1-26}) * Li_26(z^2)

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
`mock_theta_q26.py`	f_26(q), phi_26(q), psi_26(q) q-series	Proper q-Pochhammer (a;q)_n

Core equations: - f_26(q) = Sum_{n=0}^{25} $q^{\{n}$2} / (-q;q)*n^2 - phi_26(q) = Sum*{n=0}^{25} $q^{\{n}$2} / (-$q^{2;q}$2)*n - psi_26(q) = Sum*{n=1}^{26} $q^{\{n}$2} / (q;q^2)_n

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
`ramanujan_pi_uqff.py`	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
`mock_theta_pi_wstp_kernel.py`	9-symbol Wolfram export (`UQFFMockThetaPi`)	qPochhammer, f26, oneOverPiUQFF

Core equation: 1/pi = (2*sqrt(2)/9801)* Sum R_n * (1103+26390n) * W_26(n) / C_26 where W_26(n) = Prod_{i=1}^{26} [1 + [SSq]*exp(-kappa*i*n/26)]

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	5.787 x 10^-9 s^-1	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
H_SCm	0.99	SCm manifold completeness
rho_SCm	7.09 x 10^-37 kg/m^3	SCm vacuum density
rho_UA	7.09 x 10^-36 kg/m^3	UA aether vacuum density
omega_SCm	2*pi x 1.25 THz	SCm phonon resonance
sigma_0	10^-4	Base neutron cross-section

Implementation: all modules operational in `CondensedPhysics.py`, `CondensedPhysics2.py`, `MAIN_1_CoAnQi.cpp`, and Wolfram kernels (`uqff_kozima_kernel.wl`, `uqff_s26_kernel.wl`, `uqff_mock_theta_pi_kernel.wl`).